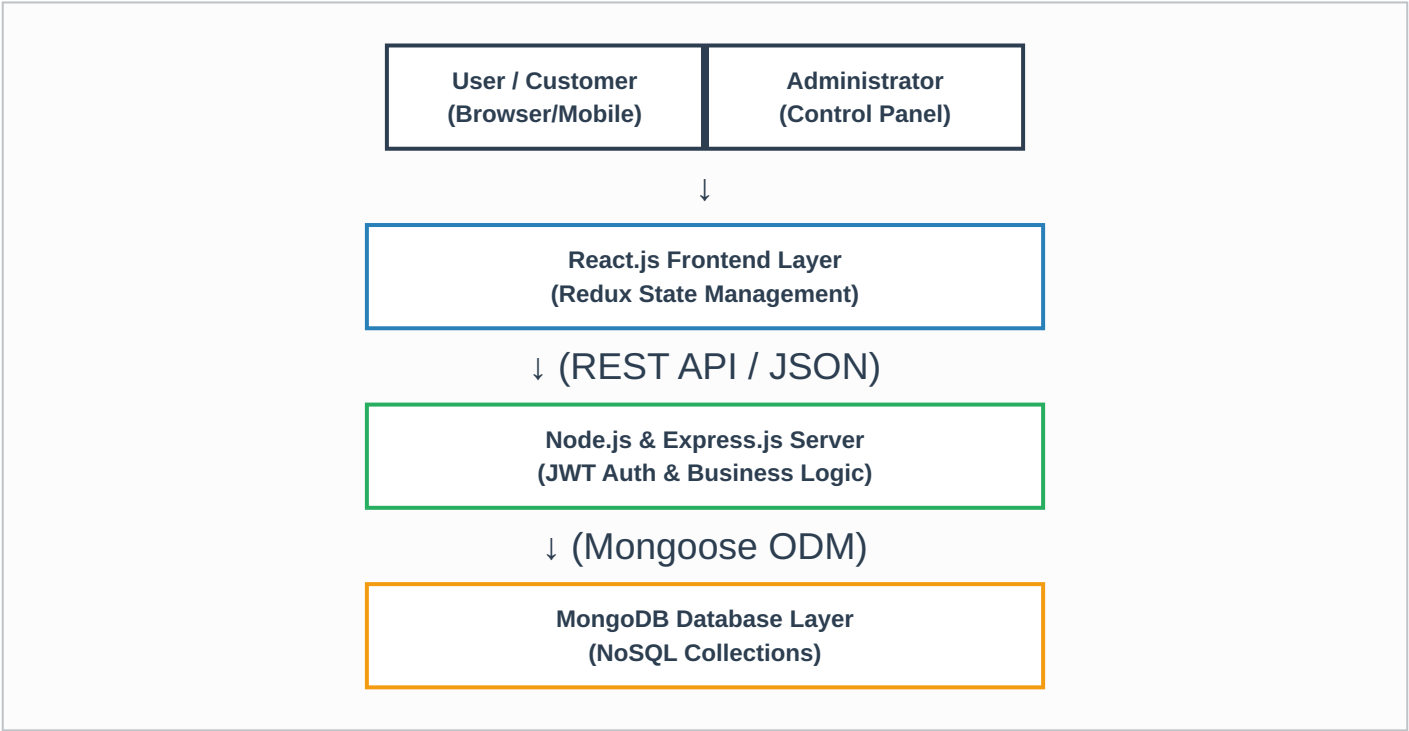


PROJECT DESIGN PHASE-II

Technology Stack & System Architecture

Date	18 June 2026	Project Name	ShopEz (MERN E-Commerce)
Team ID	--	Maximum Marks	4 Marks

1. Technical Architecture (MERN Stack)



2. Components & Technology Stack

Component	Description	Technology
Client Interface	Single Page Application (SPA) for Customers and Admin.	React.js, Tailwind CSS
Backend Server	Handles RESTful requests, Auth, and Business Logic.	Node.js & Express.js
Primary Database	Cloud-hosted document storage for users/products.	MongoDB Atlas
Authentication	Secure session management and password hashing.	JWT & Bcrypt.js
State Management	Global management of Cart and User data.	React Context API
Infrastructure	Development and Production host environment.	Local / Vercel / Heroku

3. Module Design & Data Architecture

Module	Functional Description	Key Database Fields
Auth Module	Handle user signup, login, and token generation.	email, password (hash), role
Product Module	Catalog management, category filtering, and search.	name, price, category, stock
Order Module	Transaction handling and history tracking.	userID, items[], total, status

4. API Interface Design (REST)

Endpoint	Method	Purpose
/api/users/login	POST	Authenticate user and return JWT.
/api/products	GET	Fetch all products or filtered categories.
/api/orders/place	POST	Save order details to database.
/api/admin/product	POST/PUT	Admin-only product management.

5. Design Considerations

Scalability: Using NoSQL (MongoDB) allows for flexible schema growth and horizontal scaling. **Security:** All private routes are guarded by JWT middleware; passwords never stored in plain text. **Performance:** React's Virtual DOM ensures high-speed UI updates during filtering and cart actions.